SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Case Study Analysis

Course Code:	CSA65	**Course Title:**	Generative AI and Large Scale Models
CO Assessed:	CO2 – Precision (BL1, BL2)	**Assessment:**	Assessment Tool 2 – Case Study Analysis
Weightage:	20% of CIA	**Total Marks:**	100 (1 Question)
CO1 Statement:	*Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems.(BL1, BL2)*		
Student Name:	Ch rama Karthik reddy	**Reg. No.:**	192472281
Section / Batch:	2024	**Date:**	3/8/2026

Instructions to Students

- Answer any ONE questions. Total 100 marks
- Include architecture, explanation, suitable Python/API approach, advantages, limitations and evaluation. Time allotted: 30 mins

Code of Conduct

I _______karthik/192472281___________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

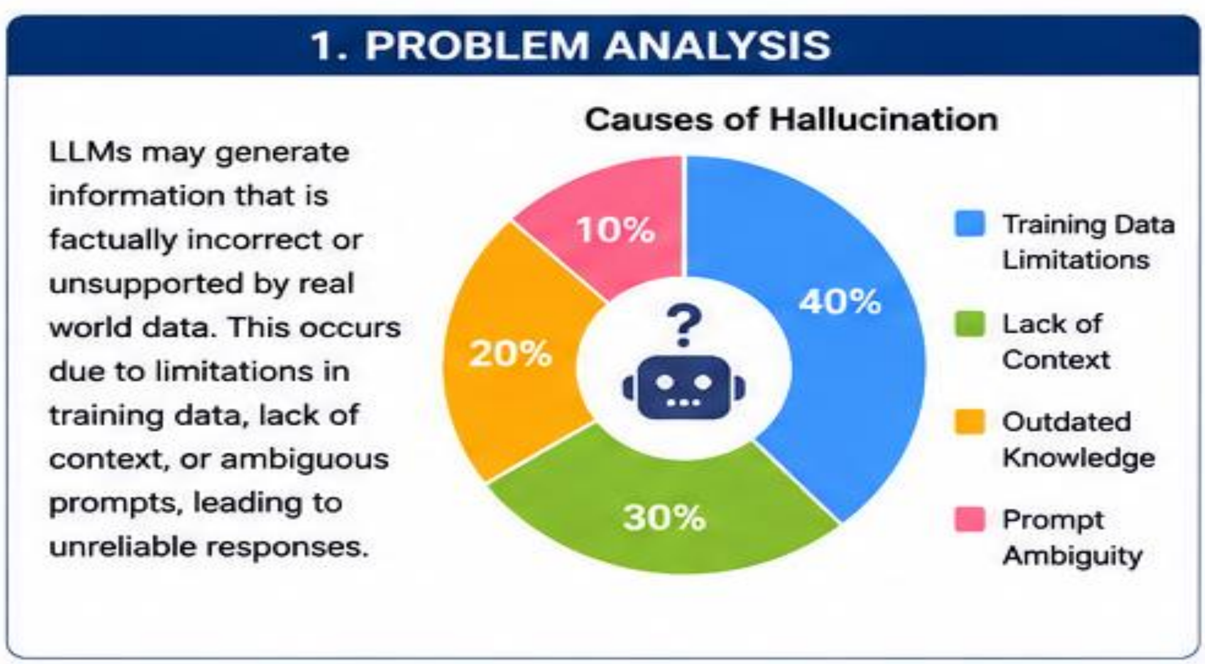
SECTION A – Case Study Analysis

Q11. Case Study 11

Develop an appropriate solution for the given Generative AI/LLM scenario. Your answer should include analysis, justification, suitable model selection (GPT/BERT/RLHF/LoRA/PEFT where applicable), implementation approach, expected outcome, and limitations.
Topic: Hallucination detection

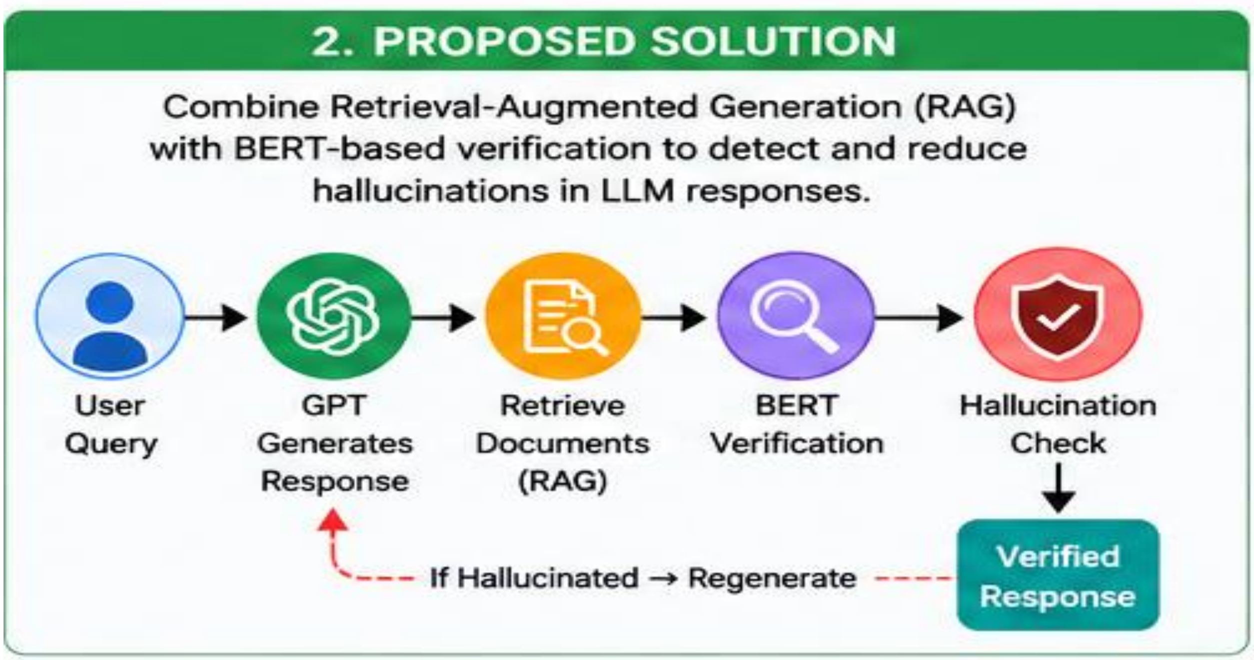
1. PROBLEM ANALYSIS

Large Language Models (LLMs) can generate responses that sound convincing but are factually incorrect or unsupported by reliable evidence. This phenomenon is known as hallucination. Hallucinations occur because LLMs generate text based on learned language patterns rather than verifying facts from trusted sources. Common causes include limitations in training data, lack of context, outdated knowledge, and ambiguous user prompts. These incorrect responses can reduce user trust and create serious risks in domains such as healthcare, finance, law, and education. Therefore, identifying and reducing hallucinations is essential for building reliable AI systems.



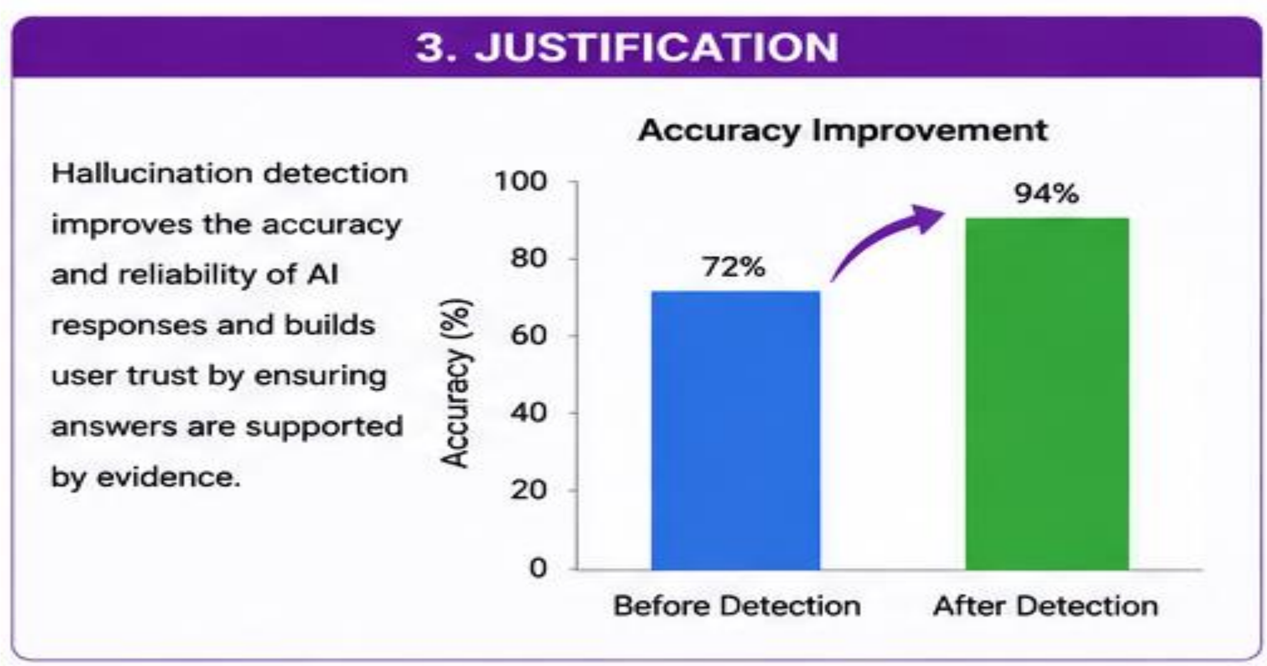
2. PROPOSED SOLUTION

The proposed solution combines **Retrieval-Augmented Generation (RAG)** with **BERT-based verification** to detect and reduce hallucinations. GPT first generates an answer to the user's query. The RAG module retrieves relevant documents from a trusted knowledge base to provide factual evidence. BERT then compares the generated response with the retrieved documents to verify semantic consistency. If hallucinations are detected, the response is regenerated until it satisfies the verification criteria. This approach produces accurate, evidence-based, and trustworthy AI responses.



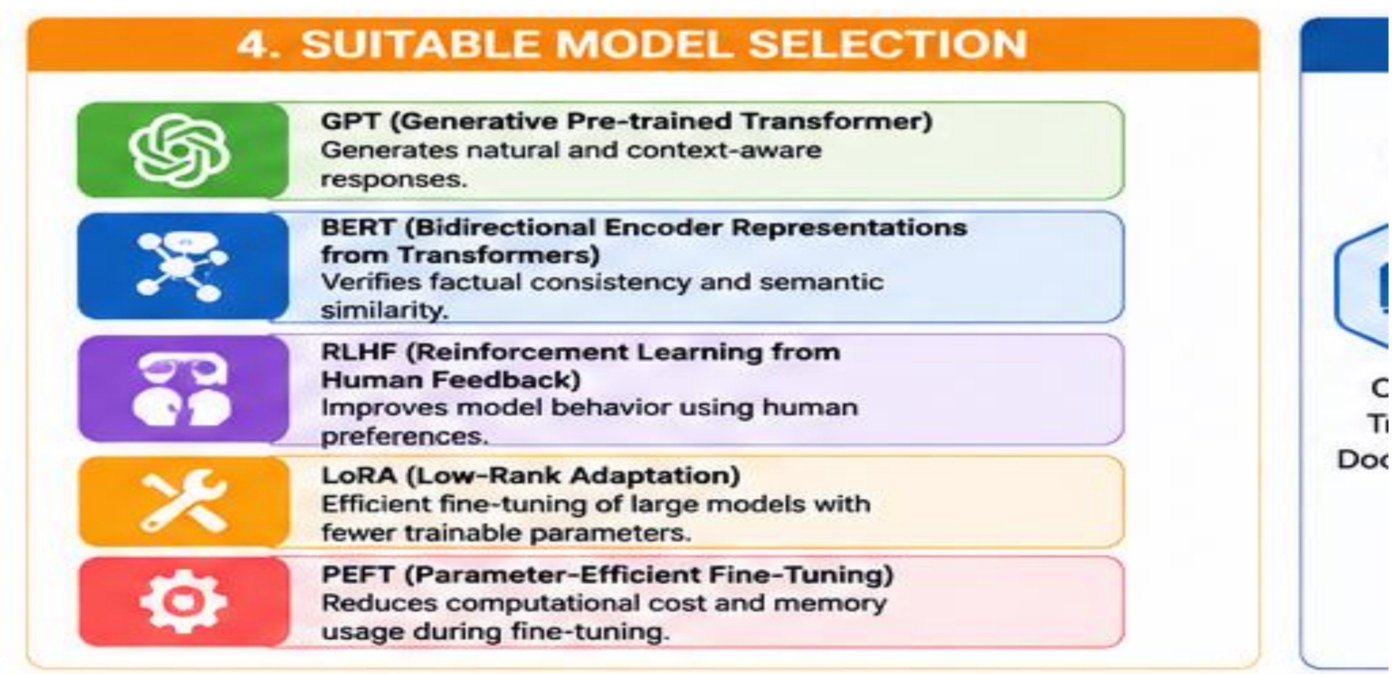
3. JUSTIFICATION

Hallucination detection improves the quality, accuracy, and reliability of AI-generated responses. By verifying answers using trusted documents, the system minimizes misinformation and ensures that responses are supported by evidence. This increases user confidence and makes AI systems more suitable for real-world applications. The accuracy improvement shown in the diagram demonstrates the effectiveness of combining GPT, RAG, and BERT for hallucination detection.



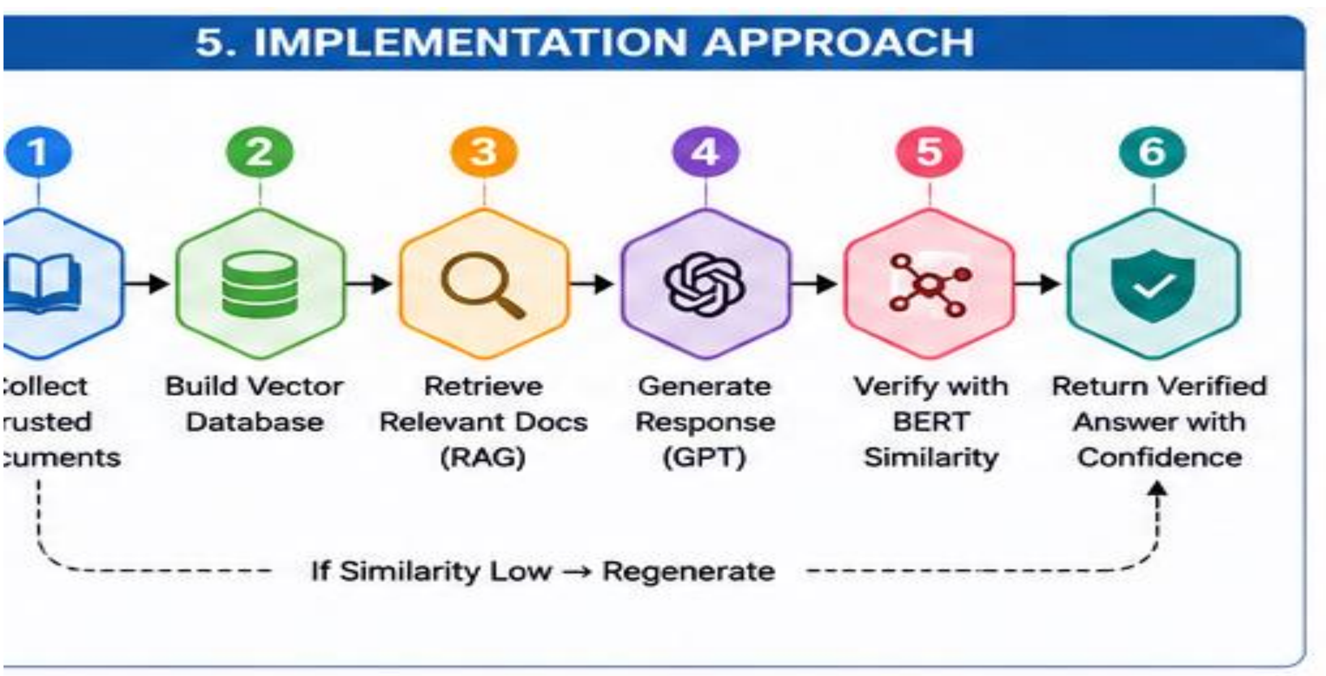
4. SUITABLE MODEL SELECTION

The proposed system combines multiple AI models and fine-tuning techniques to improve response quality. **GPT** is used to generate natural and context-aware responses. **BERT** verifies factual consistency by comparing generated answers with retrieved documents. **RLHF (Reinforcement Learning from Human Feedback)** improves model behavior using human preferences. **LoRA (Low-Rank Adaptation)** enables efficient fine-tuning with fewer trainable parameters, while **PEFT (Parameter-Efficient Fine-Tuning)** reduces computational cost and memory usage during model adaptation. Together, these techniques create a robust hallucination detection framework.



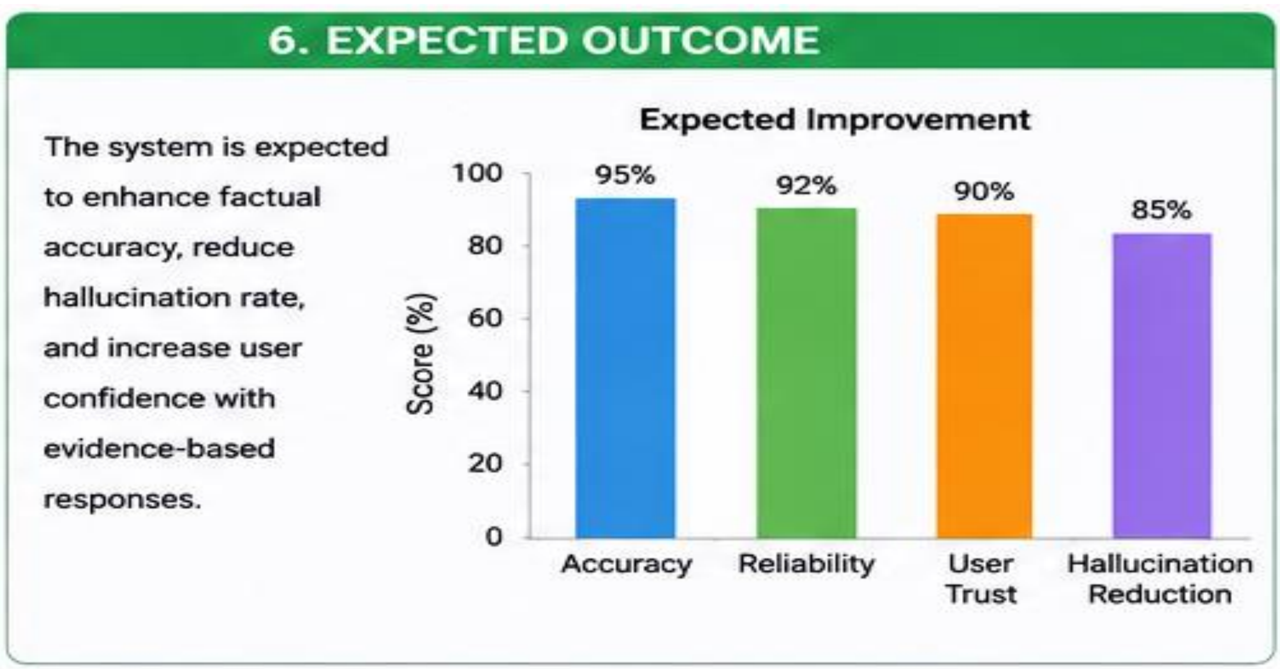
5. IMPLEMENTATION APPROACH

The implementation follows a systematic six-step process. First, trusted documents are collected from reliable sources. These documents are converted into embeddings and stored in a vector database. When a user submits a query, the RAG system retrieves the most relevant documents. GPT generates a response using both the query and retrieved information. BERT verifies the generated response by measuring semantic similarity with the retrieved evidence. If the similarity score is low, the response is regenerated; otherwise, the verified response is returned with a confidence score.



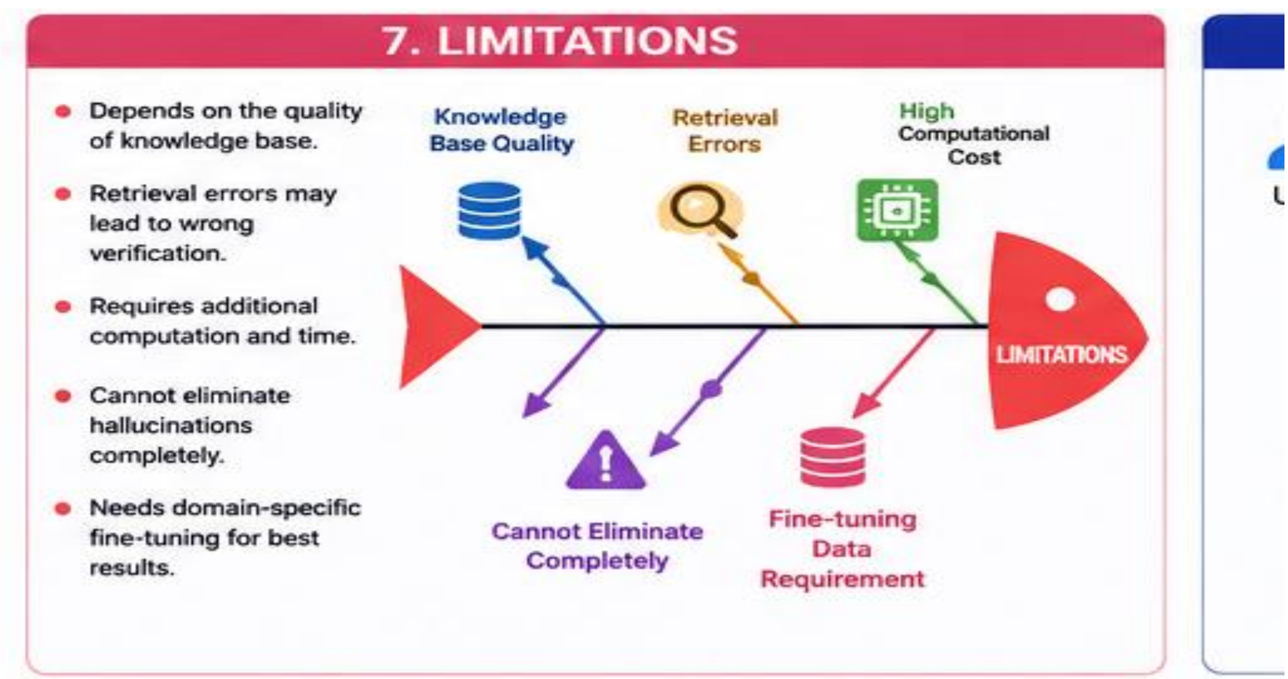
6. EXPECTED OUTCOME

The proposed system is expected to significantly improve the reliability of AI-generated responses. By verifying answers with trusted knowledge, the framework enhances factual accuracy, reduces hallucination rates, increases user trust, and provides evidence-supported responses. The system also improves decision-making in critical applications where correctness is essential. Overall, the framework enables safer and more dependable deployment of Large Language Models.



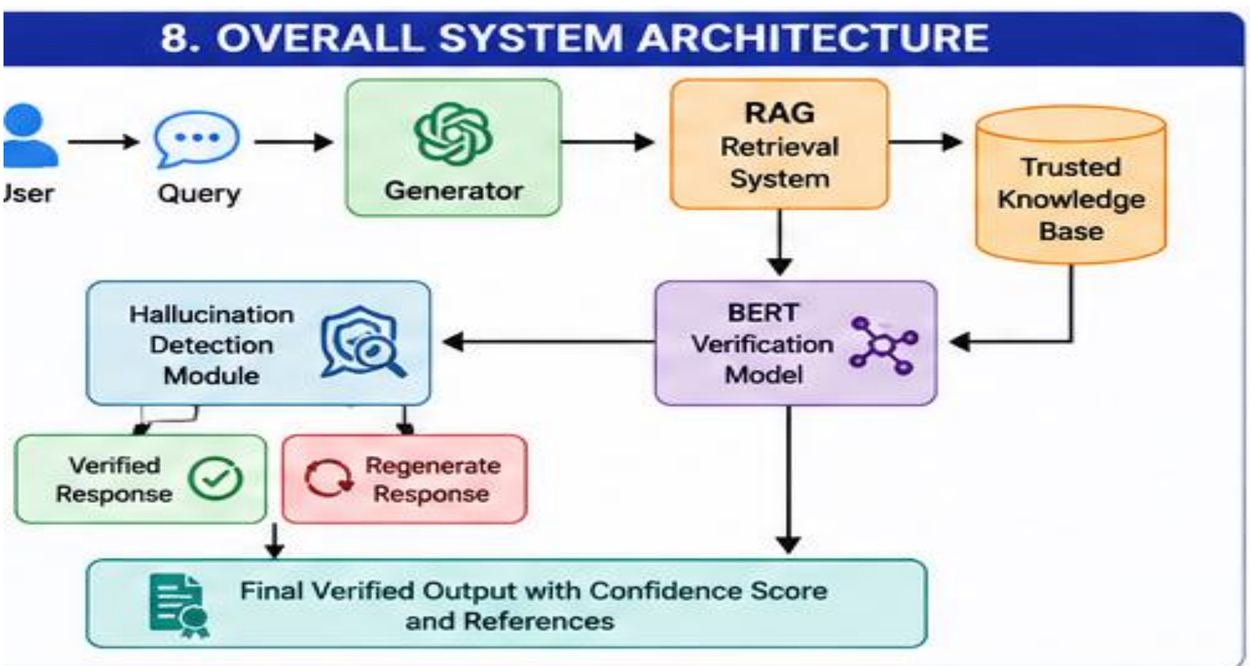
7. LIMITATIONS

Although the proposed framework effectively reduces hallucinations, several limitations remain. The system depends on the quality and completeness of the knowledge base. Retrieval errors may affect the verification process and lead to incorrect conclusions. The additional retrieval and verification steps increase computational cost and response time. Furthermore, hallucinations cannot be completely eliminated because language models still generate responses probabilistically. Domain-specific fine-tuning is also required to achieve optimal performance in specialized applications.



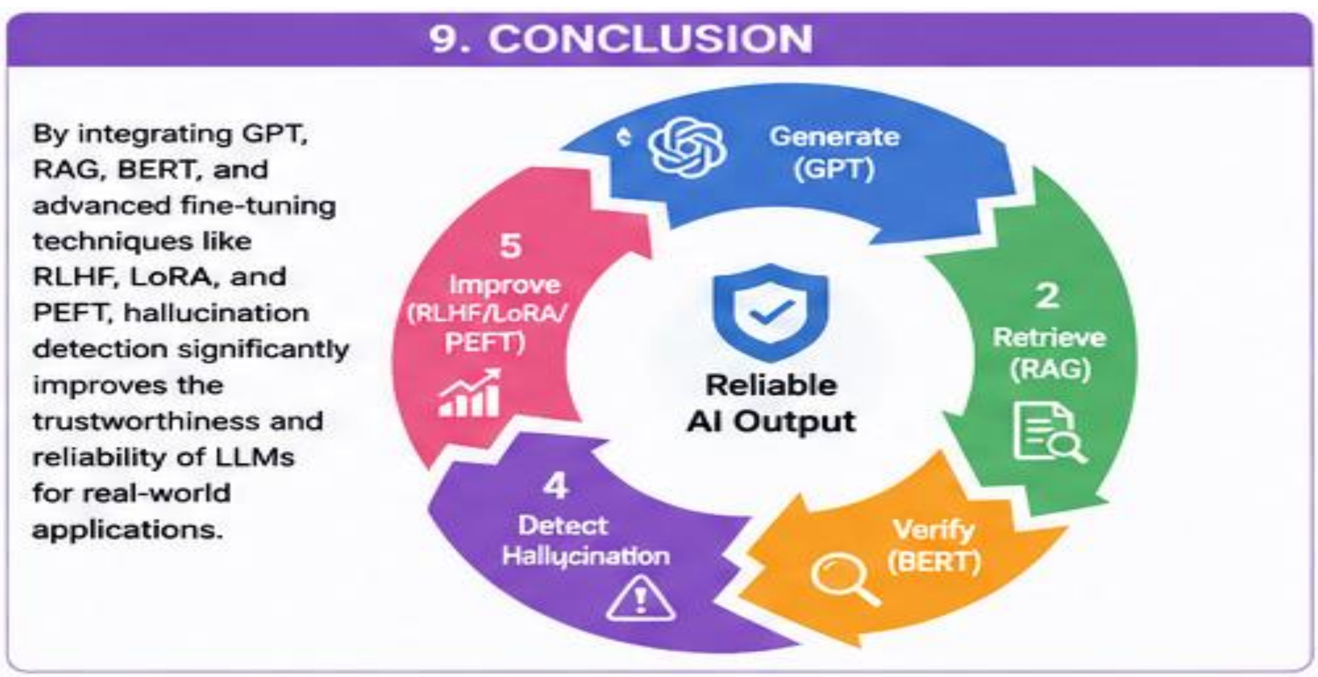
8. OVERALL SYSTEM ARCHITECTURE

The overall architecture integrates GPT, RAG, and BERT into a unified hallucination detection framework. The user query is first processed by GPT to generate an initial response. The RAG module retrieves relevant information from a trusted knowledge base. BERT then verifies the generated response by comparing it with the retrieved evidence. The hallucination detection module determines whether the response is reliable. If verified, the response is returned with a confidence score and references; otherwise, the response is regenerated until satisfactory verification is achieved.



9. CONCLUSION

Hallucination detection is essential for improving the trustworthiness of Generative AI systems. By integrating GPT for response generation, RAG for knowledge retrieval, and BERT for factual verification, the proposed framework significantly improves response accuracy and reliability. The addition of RLHF, LoRA, and PEFT further enhances model performance through efficient fine-tuning. Although challenges such as retrieval quality and computational cost remain, the proposed solution provides a practical and effective approach for deploying reliable LLMs in healthcare, finance, education, legal services, and other real-world applications.



RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Depth	30%	Deep, accurate understanding of pretraining/finetuning/RLHF concepts	Good understanding with minor gaps	Superficial understanding of concepts	Misunderstands core concepts
Real-World Implications	25%	Draws well-reasoned realworld implications and comparisons	Reasonable implications drawn	Weak or generic implications	No meaningful analysis presented
Analytical Rigor	20%	Analysis is thorough, evidence-based, and wellstructured	Mostly thorough analysis	Partial analysis with gaps in evidence	Superficial, unstructured analysis
Report Structure	15%	Report/slides professionally structured, easy to follow	Clear structure with minor issues	Structure unclear in places	Poorly structured submission
Citation & Academic Writing	10%	Properly cites sources; clear academic writing	Mostly cited, minor lapses	Inconsistent citation practice	No citation or referencing

Marks Evaluation:

Question No.	Conceptual Depth (30%)	Real-World Implications (25%)	Analytical Rigor (20%)	Report Structure (15%)	Citation & Academic Writing (10%)	100
Q1						
					Overall Total (100)	